**Laxmi Charitable Trust's**
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**
Department of Information Technology (B.Sc.I.T Semester V)
MERN Practical

# Practical – VIII

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|---|---|
| Class: TYIT | Date/Time of Submission: 23/09/2026 |

**Aim:** API Creation and Integration

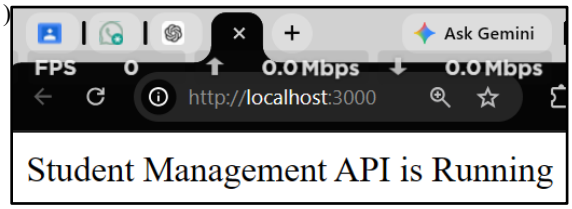
**8.A).** Create a simple REST API using Express

**SOURCE CODE:-**

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
app.use(cors());
app.use(express.json());
mongoose.connect("mongodb://127.0.0.1:27017/CollegeD
B")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
app.listen(PORT, () => {
  console.log(`Server is running on
http://localhost:${PORT}`);
});
```

***OUTPUT:-***

```
PS D:\Users\Lavkush\Study\MERN\P8> node server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```



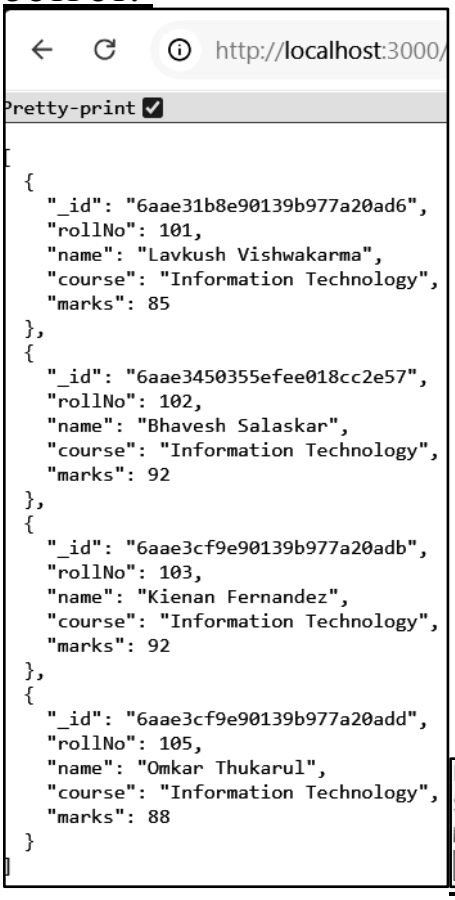

**8.B).** Implement a GET API to retrieve data.

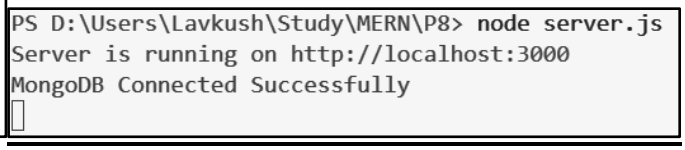
**SOURCE CODE:-**

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose
  .connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {
    console.log("MongoDB Connected Successfully");
  })
  .catch((error) => {
    console.log("MongoDB Connection Failed");
    console.log(error);
  });
// Home Route
app.get("/", (req, res) => {
  res.send("Student Management API is Running");
});
// GET API - Retrieve Students
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  } catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
// POST API - Add Student
app.post("/students", async (req, res) => {
  try {
    const student = req.body;
    await mongoose.connection.db
      .collection("students")
      .insertOne(student);
    res.json({
      message: "Student added successfully",
      student: student
    });
  } catch (error) {
    res.status(500).json({
      message: "Error adding student"
    });
  }
```

```
});
// Start Server
app.listen(PORT, () => {
    console.log(`Server is running on http://localhost:${PORT}`);
});
```

***OUTPUT:-***




**8.C).** Fetch data from the API in a React application.

**SOURCE CODE:-**

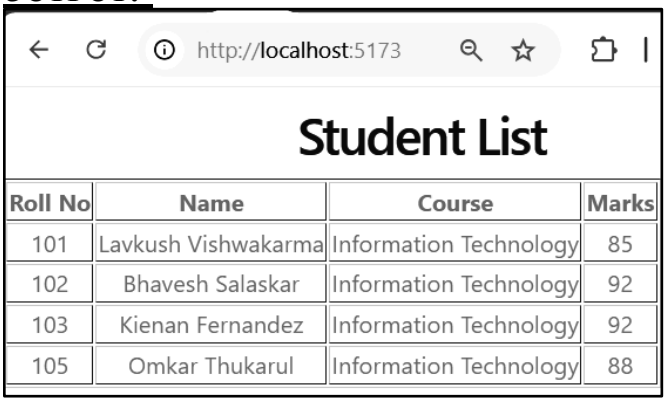
**App.jsx**

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
   const [students, setStudents] = useState([]);
   useEffect(() => {

      axios
         .get("http://localhost:3000/students")
         .then((response) => {
            setStudents(response.data);
         })
         .catch((error) => {
            console.log(error);
         });

   }, []);
   return (
      <div>
         <h1>Student List</h1>
         <table border="1">
            <thead>
               <tr>
                  <th>Roll No</th>
                  <th>Name</th>
                  <th>Course</th>
                  <th>Marks</th>
               </tr>
            </thead>
            <tbody>
               {students.map((student) => (
                  <tr key={student._id}>
                     <td>{student.rollNo}</td>
                     <td>{student.name}</td>
                     <td>{student.course}</td>
                     <td>{student.marks}</td>
                  </tr>
               ))}
            </tbody>
         </table>
      </div>
   );
}
export default App;
```

***OUTPUT:-***




**8.D).** Send data from React to the server using a POST.
**E).** Store data in MongoDB.
**F).** Display stored data in the React page.

**SOURCE CODE:-**

**App.jsx**

```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");
  // Get students
  const getStudents = () => {
    axios
      .get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };
  useEffect(() => {
    getStudents();
  }, []);
  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };
    axios
      .post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
        setName("");
        setCourse("");
        setMarks("");
        getStudents();
      })
      .catch((error) => {
        console.log(error);
      });
  };
  return (
    <div>
      <h1>Student Management System</h1>
      <h2>Add Student</h2>
      <input type="number"
        placeholder="Roll No"
        value={rollNo}
        onChange={(e) => setRollNo(e.target.value)}
      />
      <br />
      <br />
      <input type="text"
placeholder="Student Name" value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <br />
      <br />
      <input
        type="text"
        placeholder="Course"
        value={course}
        onChange={(e) => setCourse(e.target.value)}
      />
      <br />
      <br />
      <input
        type="number"
        placeholder="Marks"
        value={marks}
        onChange={(e) => setMarks(e.target.value)}
      />
      <br />
      <br />
      <button onClick={addStudent}>
        Add Student
      </button>
```

```
      <hr />
      <h2>Student List</h2>
      <table border="1">
        <thead>
          <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Course</th>
            <th>Marks</th>
          </tr>
        </thead>
        <tbody>
          {students.map((student) => (
            <tr key={student._id}>
              <td>{student.rollNo}</td>
              <td>{student.name}</td>
              <td>{student.course}</td>
              <td>{student.marks}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>  );}
export default App;
```

***OUTPUT:-***

